TYPESCRIPT / MODULE 01

ASSIGNMENT 02

Operators, Decisions & Real-World Logic

THINK. CODE. TEST.

25 multi-part programming challenges

Arithmetic / Assignment / Conditional (if-else)

10	05	10
ARITHMETIC	ASSIGNMENT	CONDITIONAL

NAME _____

BATCH _____ DATE _____

STUDENT INSTRUCTIONS

Write all programs in TypeScript using VS Code. Every main question contains related subquestions; complete all parts before moving on.

DELIVERABLE	One .ts file per main question, named Q01.ts through Q25.ts.
INPUT	For interactive problems, use prompt-sync or Node.js readline; convert numeric input using Number().
OUTPUT	Print readable labels and use template literals where helpful.
CONDITIONS	Use if / else / else if where the question requires them; avoid ternary-only answers.
VALIDATION	Check input values where the question explicitly asks you to.
SUBMISSION	Include source files and a screenshot of output for each program.

VS CODE QUICK START

```
$ node -v
$ npm -v
$ npm init -y
$ npm install -D typescript @types/node
$ npx tsc --init
$ npm install prompt-sync
$ npm install -D @types/prompt-sync
$ npx tsc
$ node dist/Q01.js
```

Run the compiled JavaScript from the output directory set in your tsconfig.json. The final command assumes your output folder is dist.

SECTION A

10 arithmetic challenges

Use +, -, *, /, %, **, and mathematical expressions.

Q01 Supermarket Checkout

A customer buys rice at Rs. 84/kg (5 kg), sugar at Rs. 48/kg (3 kg), and oil at Rs. 165/litre (2 litres).

YOUR TASKS

a. Declare meaningful variables for prices and quantities.

b. Calculate the cost of each item separately.

c. Calculate the subtotal and apply a flat Rs. 120 discount.

d. Add a delivery charge of Rs. 45 and display the final payable amount.

e. Print a formatted receipt using ONE console.log() and newline characters.

Q02 Five-Subject Report Card

A student scores 86 in Maths, 79 in English, 91 in Science, 83 in Social Science and 96 in Computer (each out of 100).

YOUR TASKS

a. Store all five marks using number variables.

b. Find the total and average marks.

c. Calculate the percentage out of 500.

d. Find how many extra marks are needed to reach a total of 450.

e. Print a report card with name, marks, total, average and percentage.

Q03 Cinema Ticket Receipt

Adult tickets cost Rs. 320 each; child tickets cost Rs. 180 each. A family buys 3 adult and 2 child tickets. A booking fee of Rs. 25 applies per ticket.

YOUR TASKS

a. Compute the total number of tickets.

b. Calculate adult and child ticket costs separately.

c. Calculate the complete booking fee.

d. Apply a flat Rs. 150 coupon to the overall booking.

e. Print the final price and the amount saved by the coupon.

Q04 Time Converter

An online training session lasts 9,785 seconds.

YOUR TASKS

a. Calculate total whole hours using Math.floor().

b. Find the remaining seconds after extracting whole hours.

c. Find the whole minutes from those remaining seconds.

d. Calculate the seconds left after extracting minutes.

e. Display the result as hours, minutes and seconds; explain why % is useful here.

Q05 Bank Account Calculator

An account begins with Rs. 25,000. A customer deposits Rs. 8,500, withdraws Rs. 6,750 and pays a service fee of Rs. 125.

YOUR TASKS

a. Declare the starting balance and all transactions.

b. Calculate the balance after the deposit.

c. Calculate the balance after the withdrawal and fee.

d. Calculate what percentage of the starting balance the withdrawal represents.

e. Determine the balance if the customer also receives Rs. 950 cashback.

Q06 Paint and Flooring Estimate

A rectangular room measures 14 metres by 11 metres. Flooring costs Rs. 625 per square metre.

YOUR TASKS

a. Calculate the area of the room.

b. Calculate its perimeter.

c. Find the flooring cost before tax.

d. Add 18% tax to obtain the final estimate.

e. Calculate the area and cost if BOTH dimensions increase by 2 metres.

Q07 Remainder Challenge

A teacher has 137 chocolates and wants to divide them equally among 12 students.

YOUR TASKS

a. Find the number of chocolates each student receives (whole number).

b. Find how many chocolates remain.

c. Find the minimum chocolates to add so everyone can receive one more.

d. Repeat the calculation with 155 chocolates and 8 students.

e. Explain in a comment how /, Math.floor() and % differ.

Q08 Electricity Bill

A household uses 275 electricity units. Rate is Rs. 6.50 per unit, fixed charge Rs. 190, and a 5% surcharge applies to usage cost only.

YOUR TASKS

a. Calculate the usage charge.

b. Calculate the surcharge on the usage charge.

c. Compute the full bill including fixed charge.

d. Calculate the bill if units increase by 40.

e. Print the difference between the original and revised bills.

Q09 Square, Cube and Powers

A square garden has side 12 metres; a cube-shaped box has side 7 centimetres.

YOUR TASKS

a. Use ** to find the area of the square.

b. Find the perimeter of the square.

c. Use ** to calculate the volume of the cube.

d. If the garden side doubles, calculate its new area.

e. Compare new area to original area as a multiplication factor.

Q10 Online Store Profit and Tax

An online store buys 15 headphones for Rs. 720 each and sells them for Rs. 980 each. Shipping costs Rs. 600 total.

YOUR TASKS

a. Calculate total purchase cost.

b. Calculate total sales revenue.

c. Find gross profit before shipping and net profit after shipping.

d. Calculate net profit as a percentage of purchase cost.

e. If two units are returned and refunded in full (no change to purchase/shipping costs), calculate revised net profit.

SECTION B

5 assignment-operator challenges

Practice =, +=, -=, *=, /= and %= by updating variables.

Q11 Wallet Transaction Tracker

A digital wallet starts with Rs. 3,000. Track sequential balance changes using assignment operators.

YOUR TASKS

a. Use += to add a cashback of Rs. 250.

b. Use -= to pay Rs. 775 for groceries.

c. Use += to add a refund of Rs. 125.

d. Use *= to double a separate rewardsPoints variable initially set to 40.

e. Print final wallet balance and reward points; never write balance = balance + ... in this exercise.

Q12 Game Score Updates

A player starts with 100 points and 3 lives.

YOUR TASKS

a. Use += to earn 75 points for completing a level.

b. Use -= to deduct 20 penalty points.

c. Use *= to apply a double-score bonus.

d. Use /= to halve the score after a special event.

e. Use %= with divisor 60 on a separate score copy and report the remainder.

Q13 Inventory Adjustments

A store initially has 75 laptops, and each one sells for Rs. 42,000.

YOUR TASKS

a. Use -= to reduce stock by 12 sold laptops.

b. Use += to receive a shipment of 20 laptops.

c. Use -= after 5 additional laptops are sold.

d. Use *= to double a promotionalPrice variable initialized to 500; explain the new value.

e. Calculate the current inventory value using the final stock and original laptop price.

Q14 Savings Growth

A student has Rs. 4,000 saved. They add or spend money over four weeks.

YOUR TASKS

a. Use += to add Rs. 850 in week 1.

b. Use -= to spend Rs. 275 in week 2.

c. Use += to add Rs. 1,200 in week 3.

d. Use *= to triple a separate bonusPoints variable initially 15.

e. Use /= to split a separate sharedExpense variable of Rs. 1,600 among 4 friends.

Q15 Counter and Remainder Lab

Start with let attempts = 0, let tasks = 24, and let tokens = 103.

YOUR TASKS

a. Increase attempts by 3 using +=.

b. Decrease tasks by 7 using -=.

c. Multiply attempts by 4 using *=.

d. Divide tasks by 2 using /=.

e. Reduce tokens to the remainder of division by 10 using %=; print all results.

SECTION C

10 conditional challenges

Build decisions with if, else if, else and input validation.

Q16 Age-Based Entry Permission

A theme park permits entry from age 12 onward. A child ticket applies to ages 12-17; adult ticket to age 18 or above.

YOUR TASKS

a. Take age as user input and convert it to number.

b. Use if/else to determine whether entry is allowed.

c. Use if/else if/else to classify child vs adult ticket, or denied entry.

d. Show a clear message for ages below 0 or above 120 as invalid.

e. Test with ages 10, 12, 17, 18 and -1.

Q17 Student Grades and Pass Rule

Five-subject total is out of 500. A student must score at least 40 in every subject to pass. Grade bands: A >= 85%, B >= 70%, C >= 55%, D >= 40%, otherwise F.

YOUR TASKS

a. Read five marks as numbers.

b. Check whether any subject mark is outside 0-100 and print Invalid marks.

c. Calculate total and percentage.

d. Use if/else to print Pass only if ALL subjects are >= 40.

e. For a passing student, assign grade A/B/C/D using an else-if ladder; otherwise F.

Q18 ATM Withdrawal Validation

An ATM account has Rs. 18,000. Withdrawals must be positive, multiples of Rs. 100, and no greater than the balance.

YOUR TASKS

a. Read withdrawal amount from the user.

b. Reject non-numeric or non-positive amounts.

c. Reject amounts that are not multiples of Rs. 100 using %.

d. Reject amounts exceeding the balance.

e. Otherwise calculate remaining balance and print Withdrawal successful.

Q19 Shopping Discount Slabs

An online store applies discounts based on order total: below Rs. 1,000 -> 0%; Rs. 1,000-2,999 -> 5%; Rs. 3,000-5,999 -> 10%; Rs. 6,000 or more -> 15%.

YOUR TASKS

a. Read the order total.

b. Reject negative or non-numeric order totals.

c. Use if/else if/else to choose the discount percentage.

d. Calculate the discount amount and payable total.

e. Test boundary values 999, 1000, 2999, 3000, 5999 and 6000.

Q20 Login Attempt Simulator

Correct username is "admin" and correct password is "Demo@123". Use fictional credentials only.

YOUR TASKS

a. Read username and password from the user.

b. If both match exactly, print Login successful.

c. If username is wrong, print Unknown username.

d. If username is right but password is wrong, print Incorrect password.

e. Add a boolean variable isAccountActive; login must fail if it is false.

Q21 Electricity Tariff Slabs

Calculate a simple electricity bill using one rate on ALL units: 0-100 -> Rs. 3/unit, 101-200 -> Rs. 5/unit, above 200 -> Rs. 7/unit.

YOUR TASKS

a. Read units consumed.

b. Reject negative units and non-numeric input.

c. Use if/else if/else to select the applicable rate.

d. Calculate the bill and add a fixed Rs. 80 service charge.

e. Print the applicable rate and total for 100, 101, 200 and 201 units.

Q22 Largest and Smallest Numbers

The user enters three different or equal numbers.

YOUR TASKS

a. Read three numbers from user input.

b. Use if/else if/else to find the largest value.

c. Use conditional logic to find the smallest value.

d. Print whether all three values are equal.

e. Test with 8, 8, 3 and with -5, -2, -8.

Q23 Leap Year Checker

A year is leap if divisible by 400, or divisible by 4 but NOT by 100.

YOUR TASKS

a. Read a year from the user.

b. Reject non-integer or non-positive years.

c. Use if/else and % to determine whether it is a leap year.

d. Test the years 1900, 2000, 2024 and 2025.

e. Explain in a comment why checking year % 4 === 0 alone is incorrect.

Q24 Delivery Charge Decision

A store delivers in two zones. Zone "local": free for orders >= Rs. 1,500, otherwise Rs. 80. Zone "remote": free for orders >= Rs. 3,000, otherwise Rs. 180.

YOUR TASKS

a. Read order amount and delivery zone.

b. Reject invalid amount or unknown delivery zone.

c. Use nested if/else statements to compute delivery fee.

d. Print subtotal, delivery fee and grand total.

e. Test local at 1499 and 1500; remote at 2999 and 3000.

Q25 Mini Payroll Eligibility Engine

An employee has a base salary and years of service. Bonus: 0-1 years -> 0%, 2-4 years -> 5%, 5-9 years -> 10%, 10+ years -> 15%.

YOUR TASKS

a. Read salary and years of service from the user.

b. Reject negative salary, negative service years or non-numeric values.

c. Select the correct bonus rate with if/else if/else.

d. Calculate bonus amount and final annual payout, assuming salary input is annual.

e. If the payout exceeds Rs. 10,00,000, print High payout; otherwise print Standard payout.